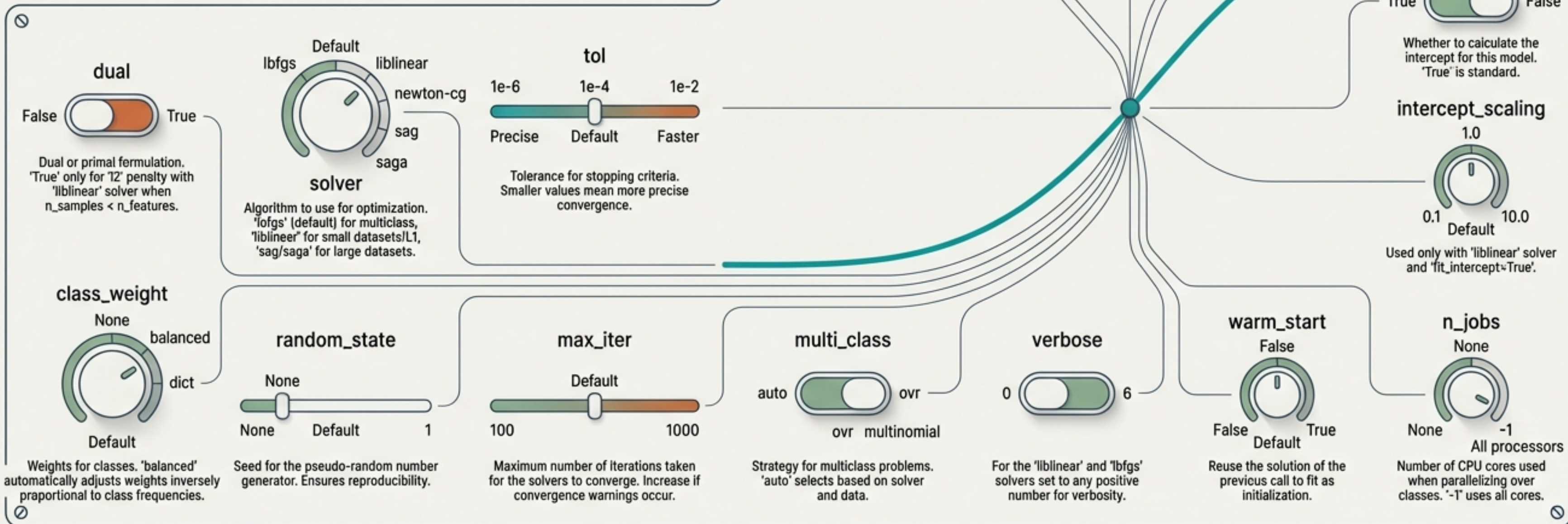


Mastering Scikit-Learn's Logistic Regression Hyperparameters

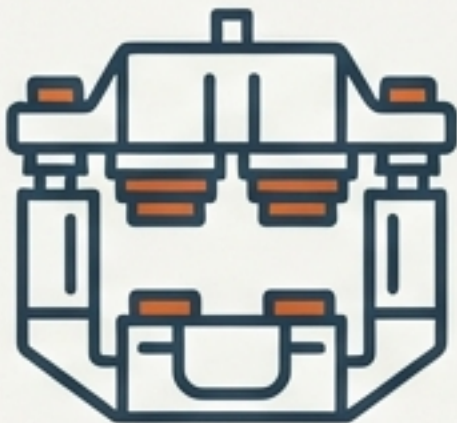
A visual cheat sheet for tuning, optimising, and controlling all 15 algorithmic parameters.



The Logistic Regression Control Room

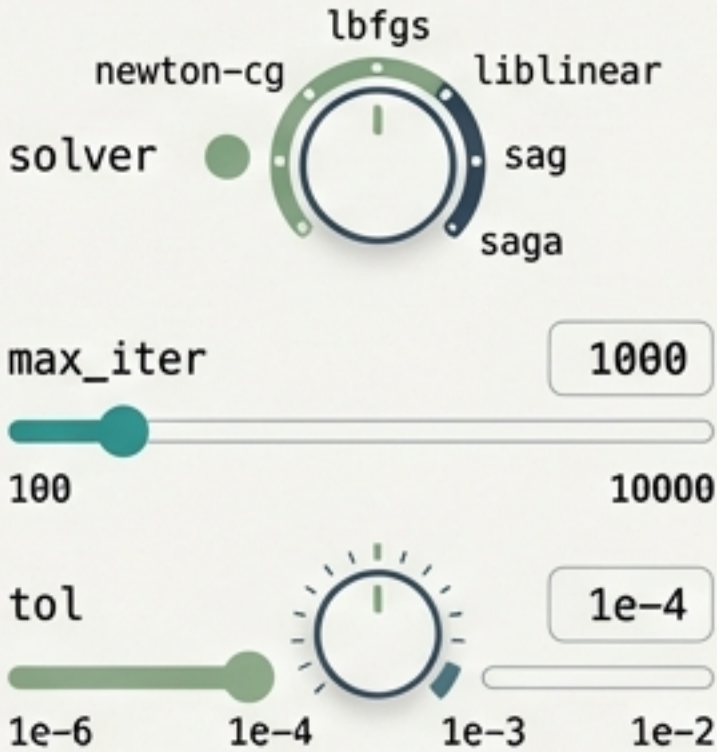
Regularisation

Controlling overfitting
Plex Sans



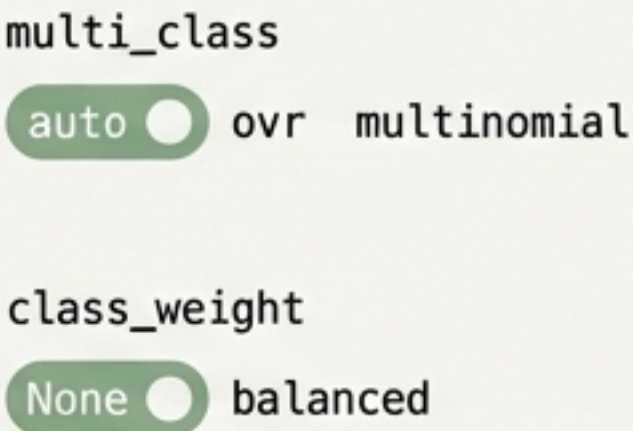
Optimisation Engine

Driving the gradient descent
Plex Sans



Data Handling

Adapting to the dataset
Plex Sans



System & Workflow

Managing hardware and reproducibility
Plex Sans



Applying Brakes with the Penalty Parameter

L2 (Default)

Ridge regularisation. Shrinks coefficients evenly.

L1

Lasso regularisation. Can shrink coefficients to zero (feature selection).

Elastic Net

Combines both L1 and L2 properties.

None

Runs standard logistic regression without any penalty applied.

The Inverse Power of C

$$C = \frac{1}{\lambda}$$

C Value

1.0 Default



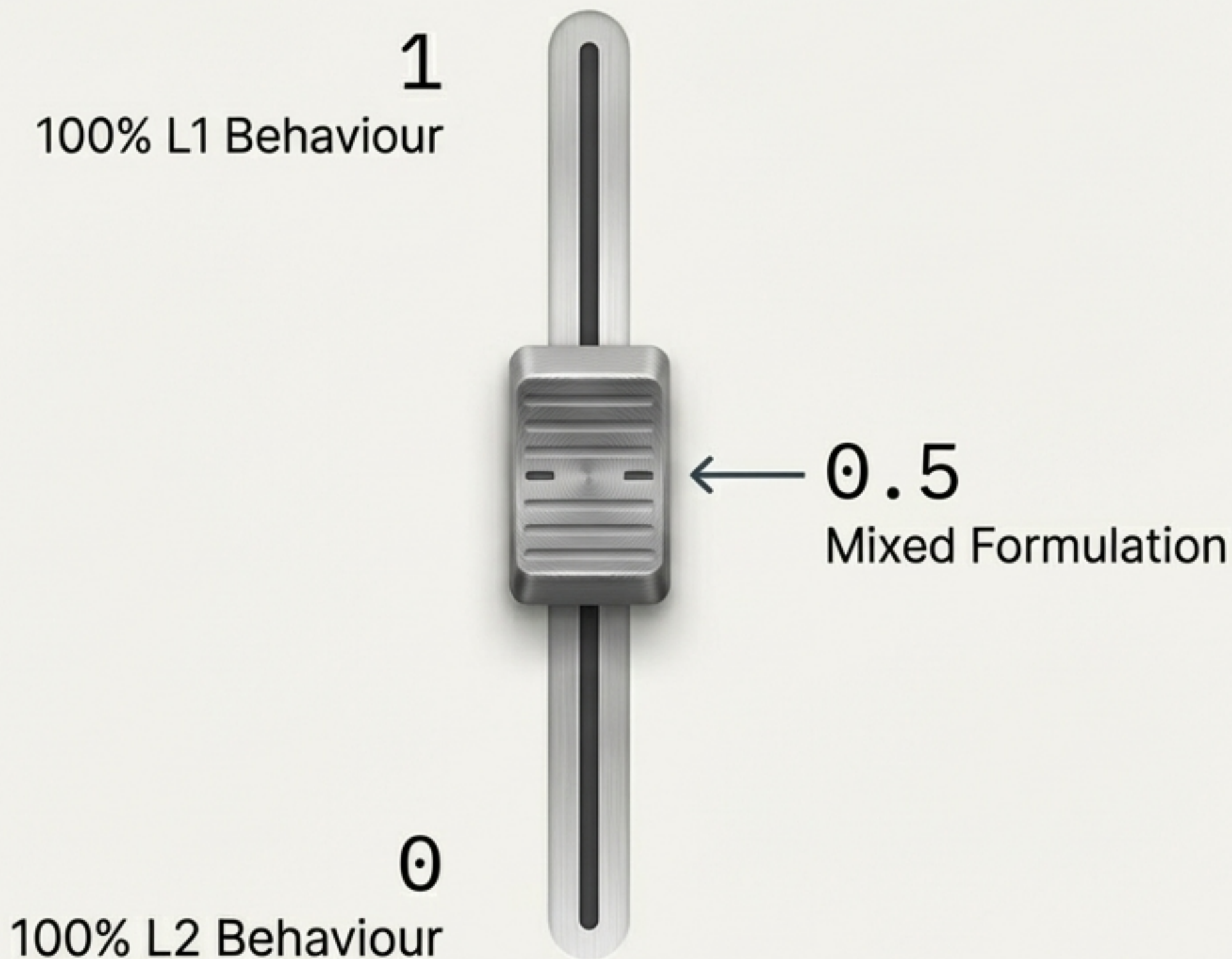
Strong Regularisation

Weak Regularisation

Pro-Tip:

C is the inverse of regularisation strength. The smaller the floating-point number, the stronger the penalty applied to the model. Moving this closer to zero aggressively flattens your decision boundary.

Blending Penalties with L1 Ratio



Condition:

This parameter is only activated when `penalty='elasticnet'`.

Function:

Determines the exact mix of L1 and L2 regularisation applied to the model.

Selecting the Optimisation Engine



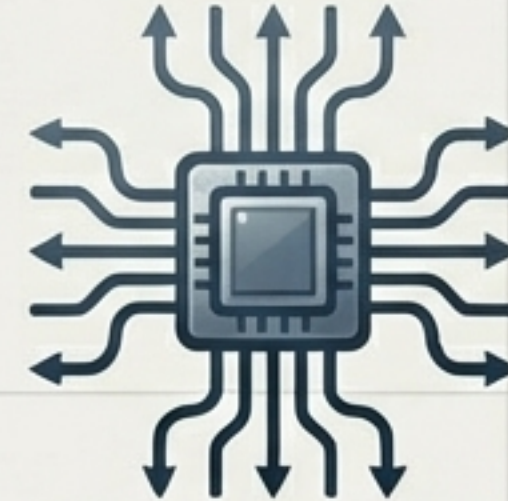
Newton-CG



LBFGS



Liblinear



SAG



SAGA

Scikit-Learn implements different optimisation techniques created by different researchers.

Consultant Note

Solvers vary in how they handle large datasets, unscaled data, and multi-class problems. However, their most critical constraint is penalty compatibility.

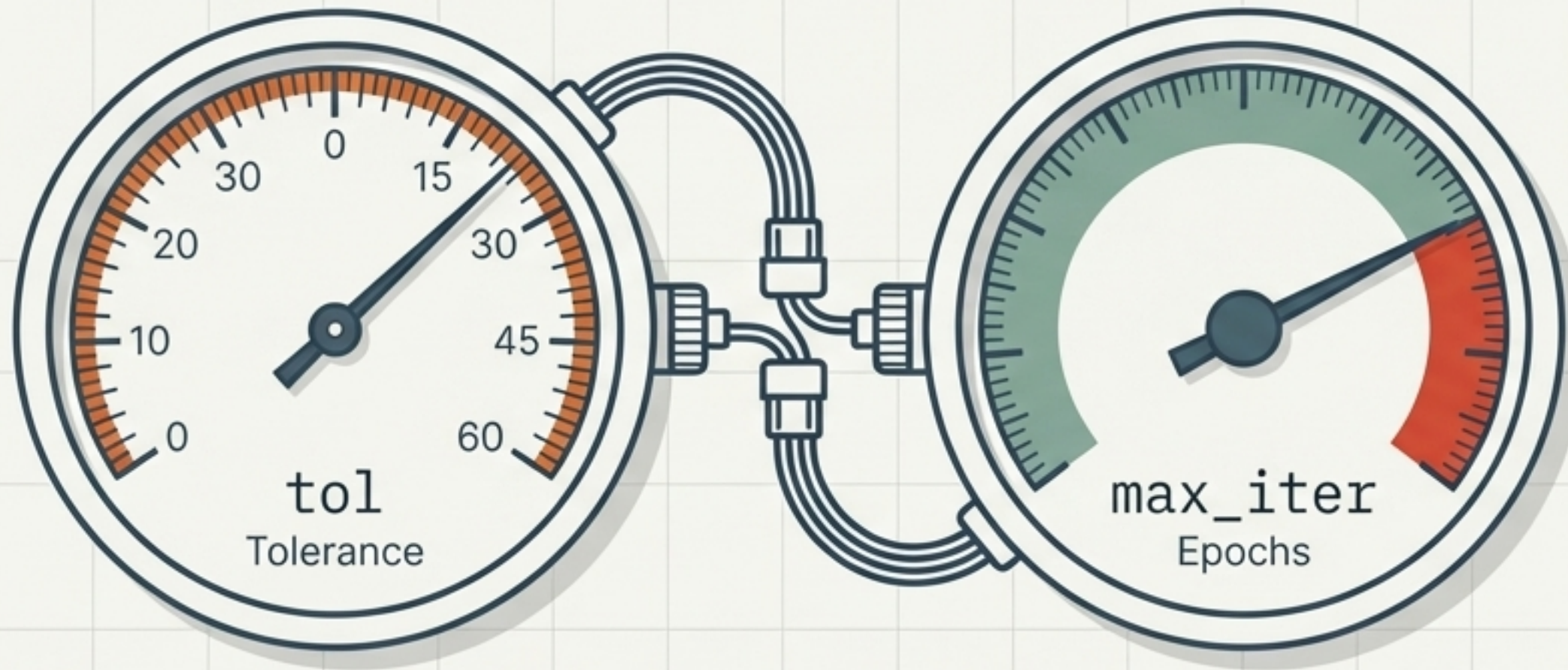
The Solver Compatibility Matrix

	L2	L1	Elastic Net	None
Newton-CG	✓	✗	✗	✓
LBFGS	✓	✗	✗	✓
Liblinear	✓	✓	✗	✗
SAG	✓	✗	✗	✓
SAGA	✓	✓	✓	✓

Highlight Note

SAGA is the only solver that supports **Elastic Net**. **Liblinear** cannot run without a penalty.

Controlling Algorithmic Convergence



`tol`

The stopping criteria. If the improvement is smaller than this value, the algorithm stops.

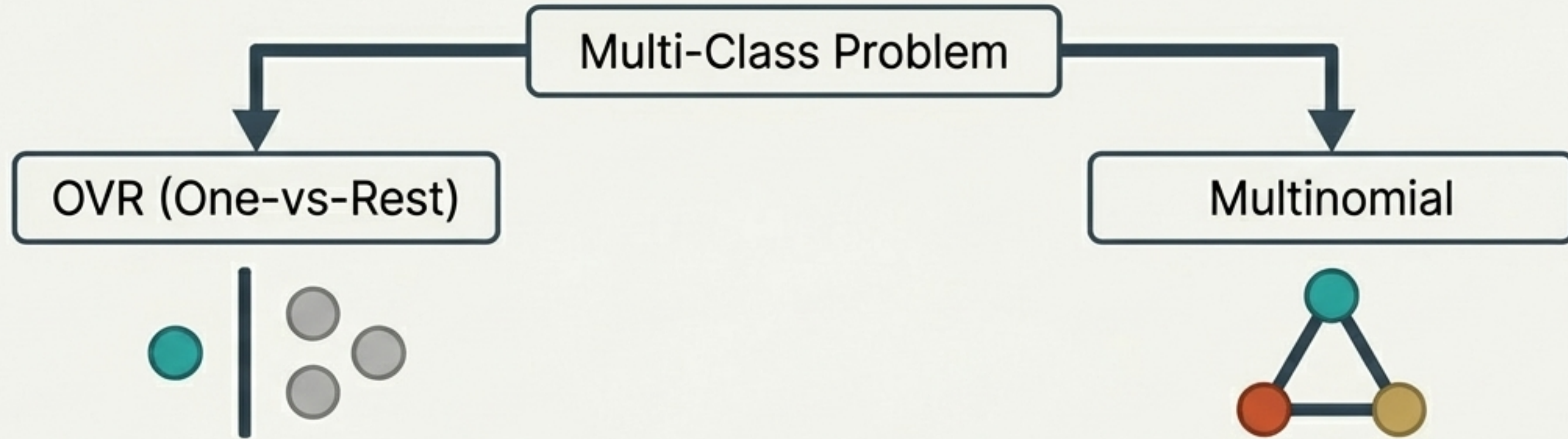
`max_iter`

The absolute maximum number of epochs the algorithm will run to reach convergence.

Pro-Tip

In practice, leave `tol` at its default. If your model isn't converging, manually increase `max_iter` rather than shrinking the tolerance.

Tackling Multi-Class Classifications



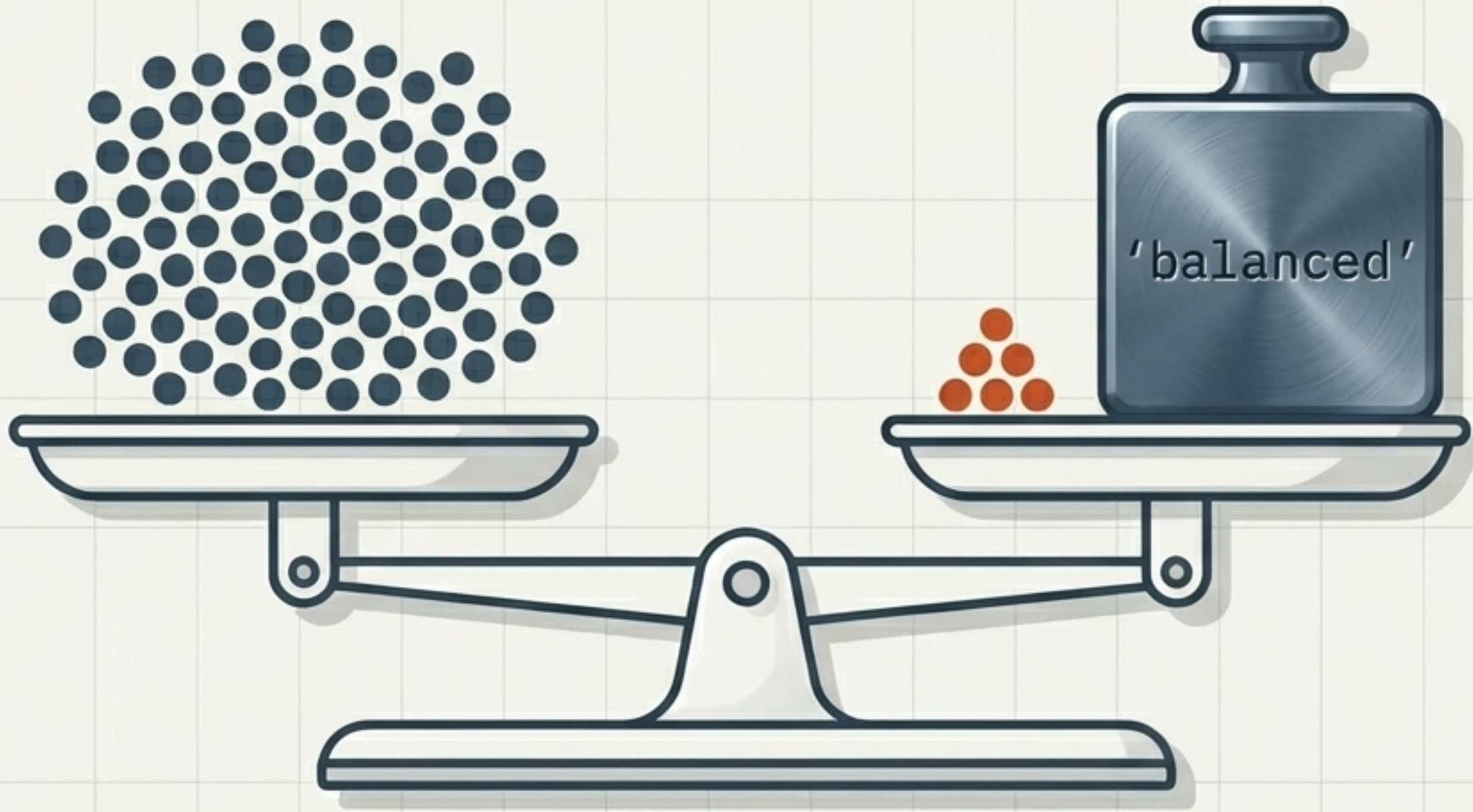
Trains a separate binary logistic regression model for every single class against the rest.
(If 3 classes = 3 models).

Minimises the multinomial loss directly across all classes simultaneously.

Auto (Default):

Automatically selects the best approach based on the data and solver. Unless you have a specific mathematical reason, leave this on 'auto'.

Rebalancing Skewed Datasets



Default: None

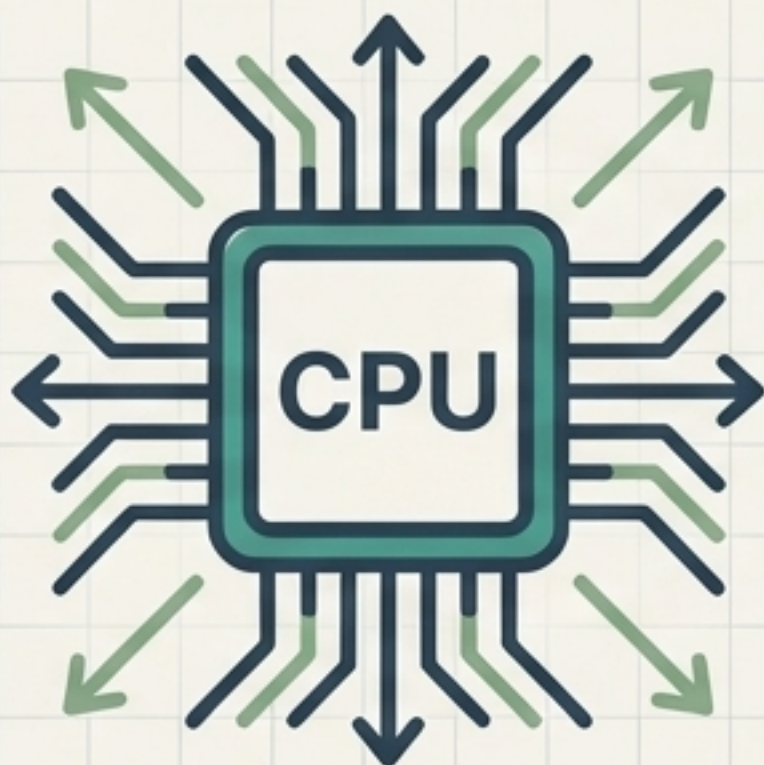
Assumes all classes have equal weight.

The Fix:

Set to 'balanced'.

Mechanism: Automatically adjusts weights inversely proportional to class frequencies. Crucial when one class heavily dominates the other in your training data.

Optimising Compute Resources



`n_jobs`

Determines how many CPU cores to utilise.

Tip: Set `n_jobs = -1` to use all available cores. This drastically speeds up computationally expensive tasks like **OVR multi-class training**.

`warm_start`

Set to **True** to reuse the solution of the previous call to fit as initialisation. Great for pausing and **continuing training** without starting from scratch.

Ensuring Visibility and Reproducibility

Terminal Output

```
Epoch 1/100, Loss: 0.854, Accuracy: 65.2%  
Epoch 2/100, Loss: 0.712, Accuracy: 72.8%  
Epoch 3/100, Loss: 0.634, Accuracy: 77.5%  
...  
Epoch 10/100, Loss: 0.410, Accuracy: 85.1%  
...
```



`verbose`

Set to 1 (or higher) to output logs and monitor the training progress of the algorithm in real-time.

`random_state`

Pass an integer value to lock the random seed.

Note: This is only utilised when the solver is set to SAG, SAGA, or Liblinear.

Under the Hood Tweaks

`fit_intercept`

Controls the W0 (bias) term. Keep as **True** unless data is already centered.

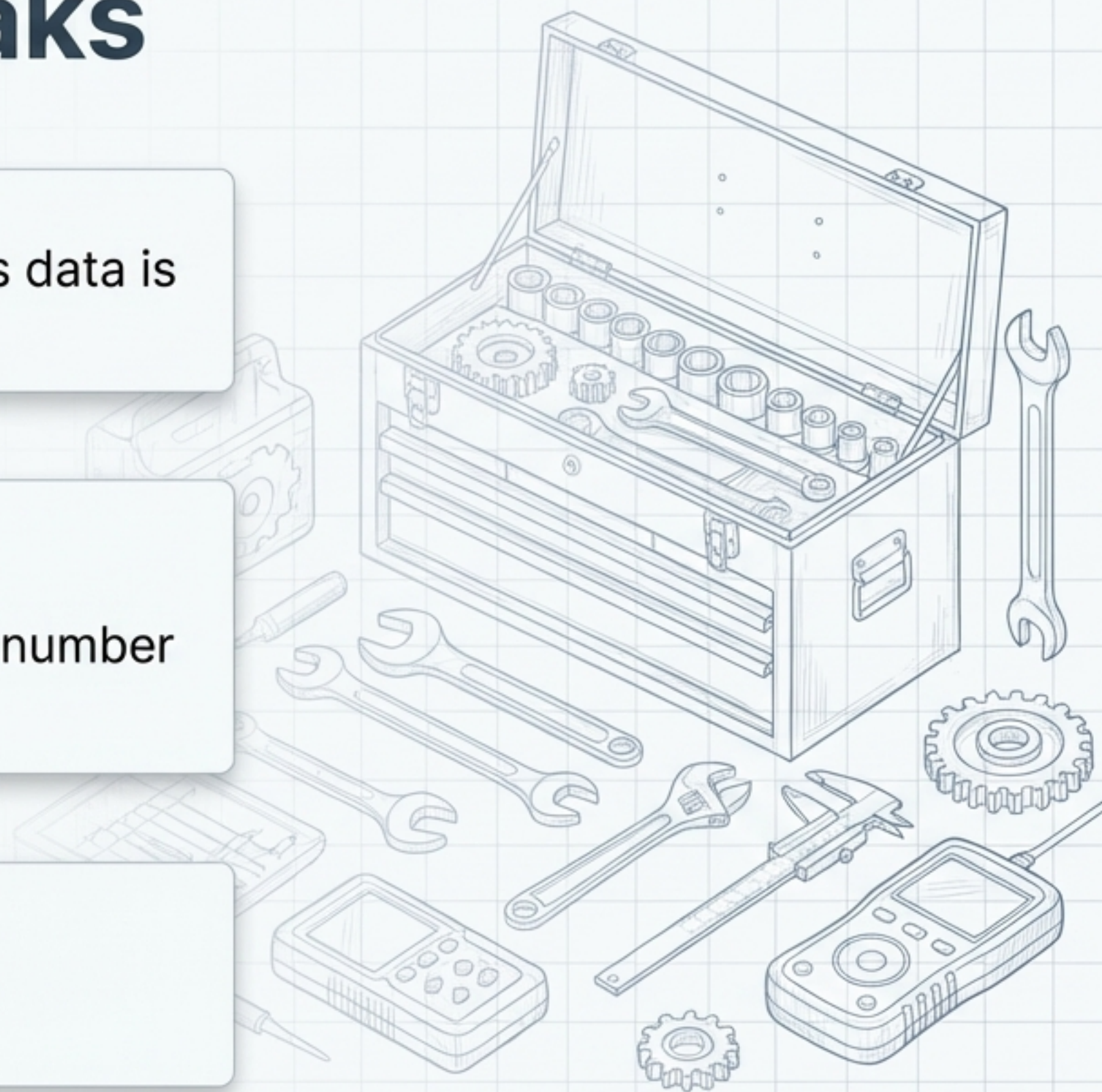
`dual`

Uses dual formulation.

Consultant Note: Keep this as **False** unless your number of rows is smaller than your number of columns.

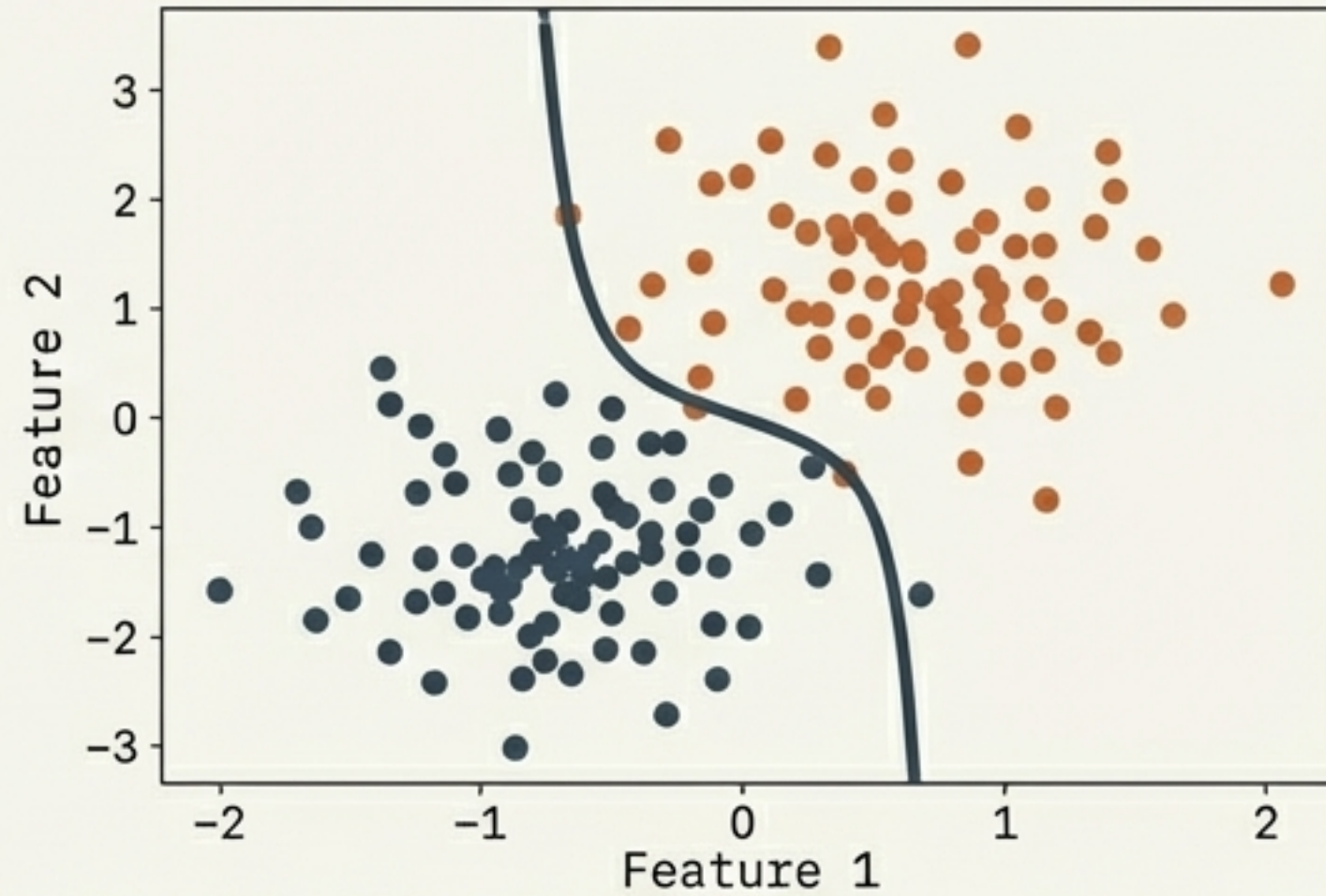
`intercept_scaling`

A highly technical parameter for Liblinear.
Generally untouched in standard workflows.

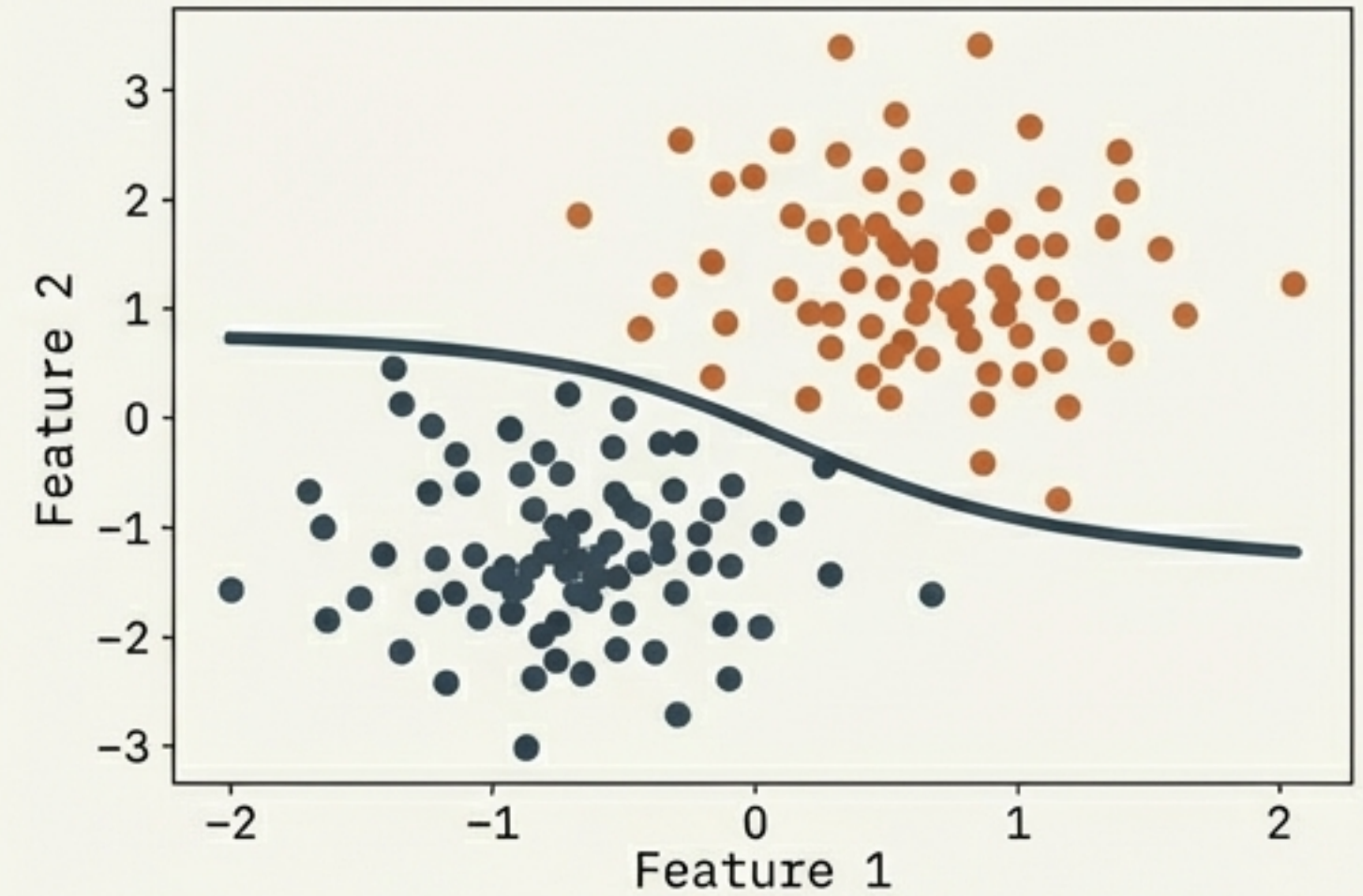


Visualising the Impact

Graphic A (High C / No Penalty)



Graphic B (Low C / Strong Penalty)



Experimentation reveals the physical mechanics of the algorithm.

Applying stronger regularisation (decreasing C) visibly reduces the slope's magnitude, preventing the model from over-fitting to the training data.

The Laboratory: Next Steps

Call to Action: The best way to master these 15 parameters is through applied experimentation.



Iris

Perfect for testing
Multi-class (`OVR` vs
`Multinomial`).



Breast Cancer

Ideal for tuning `C`,
`Regularisation`, and
`class_weight`.



MNIST

Test computational limits
by tuning `n_jobs` and
testing the `saga` solver on
large-scale data.